Experiment - 02

1) P-A.

AND, OR, XOR & Nibble making

Gn

X = A7H = 1010 0111

Y = 3CH = 0011 1100

W = 1234H.

AND

X = 1010 0111

Y = 0011 1100

0010 0100

X + Y = 24H

OR

X = A7H = 1010 0111

Y = 3CH = 0011 1100

1011 1111

X OR Y = BFH.

XOR :-

X = 1010 0111

Y = 0011 1100

1001 1011

X XOR Y = 9BH

Lower Nipple Masking.

Mask = 0FH = 0000 1111

X = 1010 0111

Mask = 0000 1111

0000 0111

LNM = 07H

Upper Nibble Masking.

Mask = F0H = 1111 0000

X = 1010 0111

Mask = 1011 0000

1010 0000

UNM = A0H.

One's Complement of X:

X = A7H = 1010 0111

X' = 0101 1000

Result : 58H.

Two Complement of X

X' = 0101 1000

+ 1

0101 0100

Result = 59H.

∴ A7H + 59H = 100H

8) One's Complement of 16-bit W

W = 1234 H

∴ W = 0001 0010 0011 0100

W' = 1110 1101 1100 1011

Result : EDCBH

9) Two's complement of 16-bit W

One's Complement = EDCBH

+ 0001H

EDCCH

Verification:-

1234H + EDCCH = 10000H

10) Flag Calculation After AND

Result = 24H = 0010 0100

S = 0, Z = 0, AC = 1, P = 1, CY = 0

Program - B

Shift & Rotate operation.

Gn

D = CAH = 1100 1010

W = 9234H.

1) Set Bits 7 & 0 ; Mask = 81H.

D = 1100 1010

Mask = 1000 0001

1100 1011 ∴ CBH.

Clear Bits 3 & 1

Mask = F5H

D = 1100 1010

Mask = 1111 1010

1100 0000 ⇒ result = C0H.

Toggle lower Nibble :

Mask = 0FH

D = 1100 1010

Mask = 0000 1111

1100 0101 ⇒ result = C5H.

Test Bit 6 :-

Mask = 40H

D = 1100 1010

Mask = 0100 0000

0100 0000 ⇒ result = 40H

Result is ~~stored~~ as 01H.

Rotate left (RLC)

D = 1100 1010

Ori = 1100 1010

Rot = 1001 0101 ∴ => 95H. CY = 1

Rotate right (RRC)

Ori = 1100 1010

Rot = 0110 0101

Result = 65H ; CY = 0

Logical Shift left

Ori = 1100 1010

Shifted = 1001 0100

result = 94H ; CY = 1

Logical Shift right:

Ori = 1100 1010

Shifted = 0110 0101

result = 65H ; CY = 0

Arithmetic Shift right:

Ori = 1100 100

Shifted = 1110 0101

Result = E5H ; CY = 0

Signed interpretation = - 54 ÷ 2

= - 27

Count of no. 1's

D = 1100 1010

∴ No. of 1's = 4

16 bit Shift left.

Qn:

W = 9234 H.

9234 = 1001 0010 0011 0100

Shift = 0010 0100 0110 1000

Result = 2468H ; CY = 1